National University of Computer & Emerging Sciences
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Edge and IoT Networking

Simulation

Project Proposal
Computer Networks
Section: 5J

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Project Overview

This project focuses on simulating a small-scale Edge and IoT network that demonstrates how data generated by IoT devices is routed from the edge layer to the cloud for processing. The simulation models how multiple devices communicate efficiently with limited bandwidth while maintaining reliable data transfer. It highlights essential networking concepts such as data routing, bandwidth optimization, and device registration in an IoT ecosystem.

Project Description

In this project, multiple simulated IoT devices (e.g., sensors) will generate data readings and send them to an Edge Node using socket-based communication.

The Edge Node acts as an intermediate processing point — performing simple data filtering, aggregation, or compression — and then forwards processed data to the Cloud Server.

This setup will mimic real-world IoT architectures where data is first processed near the source (edge computing) to reduce latency and network congestion before reaching the cloud for long-term storage or analytics.

The entire system will be implemented locally using Python's networking libraries without any GUI or third-party cloud services.

Functional Requirements

1. Device Registration: Each IoT device must register itself with the edge server before transmitting data.
2. Data Transmission: Devices periodically send simulated sensor data to the edge node.
3. Edge Processing: The edge node aggregates or filters incoming data (e.g., averages temperature readings).
4. Cloud Forwarding: The edge node forwards processed data packets to the cloud server.
5. Bandwidth Optimization (Optional): Implement simple rate limiting or compression simulation.
6. Logging: Maintain logs of all communications and data transfers for verification.

Key Features

- Edge-to-Cloud Data Routing Simulation using sockets.
- Multiple IoT Devices simulated via threads or processes.
- Local Execution Only – no external cloud dependency.
- Data Aggregation at Edge to mimic bandwidth optimization.
- Real-Time Logging of communication events and packet flow.

Technical Specifications

- Communication Model: Client-Server using TCP sockets.
- Architecture Layers:

  IoT Device Layer → Edge Node → Cloud Server

- Data Format: JSON strings or simple delimited text messages.

- Performance Metrics (Optional): Latency (send-receive delay), packet count, and message size.

Tech Stack

Component	Technology
Programming Language	Python 3
Networking	Python socket and threading modules
Data Serialization	JSON
Logging	Python logging module
(Optional) Visualization	Matplotlib for latency/bandwidth charts

Challenges & Risks

- Concurrency Management: Handling multiple IoT devices simultaneously using threads or asynchronous I/O.
- Data Synchronization: Ensuring correct ordering and integrity of packets between edge and cloud layers.
- Scalability Limits: The local simulation may face performance issues if too many devices are simulated simultaneously.
- Bandwidth Simulation: Implementing realistic delay or compression effects may require careful parameter tuning.